COMS 3003A
Test 3 – Deferred version

22 May, 2024

This is a closed book test.

Time allowance: 60 minutes

Total number of points: 50

1. For each of the following formulas, find a model where the formula is true and a model where the formula is false:
   (a) **(6 points)** $\forall x\, \forall y\, (R(x, y) \to \exists z\, R(x, z) \land R(z, y))$;
   (b) **(6 points)** $\forall x \forall y \forall z\, (R(x, y) \land R(y, z) \to R(x, z)) \land \forall x \exists y R(y, x) \land \forall x\, R(x, x)$.
2. **(10 points)** Prove that the formula $\forall x\, (P(x) \to Q(c)) \to (\exists x\, P(x) \to Q(c))$ is valid.
3. **(6 points)** Prove that the formula $\forall x \exists y R(x, y) \to \forall x R(x, x)$ is not valid.

4. In the proof of undecidability of the first-order logic, we describe computations of Turing machines on the empty string ϵ. When describing a Turing machine $M = (Q, \Sigma, \Gamma, q_0, q_1, q_2, \delta)$, we use the following vocabulary (remember that every first-order language contains the equality symbol $=$):

   - the constant symbol 0 ('zero');
   - the unary function symbol S ('successor' on natural numbers);
   - for every $k \in \{0, \ldots, |Q|\}$, a unary predicate letter $Q_k(x)$ ('at step x, the machine M is in state q_k');
   - a binary predicate letter $C(x, y)$ ('at step x, the machine M scans cell y');
   - for every $k \in \{0, \ldots, |\Gamma|\}$, a binary predicate letter $S_k(x, y)$ ('at step x, cell y contains symbol s_k').

   When describing computations of Turing machines, we will be making the following assumptions:

   - the tape is one-way infinite (hence the cells can be numbered 0, 1, 2, etc.);
   - machines can move left and right (no other types of head movement are allowed);
   - the left-most cell, i.e., cell 0, always contains an end-of-tape marker, which is a special tape symbol;
   - states are always called q_0, q_1, q_2, etc.;
   - the starting state is q_0, the accepting state is q_1, and the rejecting state is q_2;
   - the tape symbols are always called s_0, s_1, s_2, etc.
   - s_0 is the blank;
   - s_1 is the end-of-tape marker.

   Write formulas saying the following, in the vocabulary listed above:

   (a) **(4 points)** At every step of the computation, M is in exactly one state.

   (b) **(4 points)** A formula describing the initial configuration of M on the empty word ϵ.

   (c) **(5 points)** Suppose that $q_i s_k \to q_j s_l L$ is an instruction from δ. Write a formula describing what happens when M executes this instruction.

5. **(9 points)** Recall that a first-order formula φ logically follows from a set Γ of first-order formulas if φ is true in every model where all members of Γ are simultaneously true. Prove that determining logical consequence is undecidable. You may assume the following problems to be undecidable:

   - self-acceptance for Turing machines;
   - halting for Turing machines;
   - halting on the empty string for Turing machines.

   You may need to do more than one reduction.